

Duals of Hilbert Spaces; Riesz Representation Theorem; Operators in Hilbert Spaces

Ahmed ALOtaibi	202153590
Abdulaziz AlMutairi	202168150
Mohammed AlGhudiyan	202266040

April 30, 2026

Objectives

- ▶ Study the adjoint of an operator
- ▶ Study (Self-adjoint, normal, unitary operators)
- ▶ Study the Spectrum of an operator.

Outline

Adjoint

Normal Operators

Adjoint

Normal Operators